

Safety data sheet

Page: 1/9

BASF Safety data sheet
Date / Revised: 19.09.2018
Product: **Trilon® M Max BioBased Gran**

Version: 2.0

(30513095/SDS_GEN_TH/EN)

Date of print 26.08.2021

1. Substance/preparation and manufacturer/supplier identification

Trilon® M Max BioBased Gran

Use: complexing agents for the chemical industry

Manufacturer/supplier:

BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Care Chemicals
Telephone: +49 621 60-44676
Telefax number: +49 621 60-6644676
E-mail address: emd-ems-ehs-masterdata@basf.com

Emergency information:

International emergency number:
Telephone: +49 180 2273-112

2. Hazard identification

Classification according to UN GHS 2009

Classification of the substance and mixture:

| No need for classification according to GHS criteria for this product.

Label elements and precautionary statement:

| The product does not require a hazard warning label in accordance with GHS criteria.

Other hazards which do not result in classification:

No specific dangers known, if the regulations/notes for storage and handling are considered.

3. Composition/information on ingredients

Chemical nature

| Alanine, N,N-bis(carboxymethyl)-, trisodium salt
CAS Number: 164462-16-2

4. First-Aid Measures

General advice:

Remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air.

On skin contact:

Wash thoroughly with soap and water.

On contact with eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:

Rinse mouth and then drink plenty of water.

Note to physician:

Symptoms: Further symptoms and / or effects are not known so far

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:

dry powder, foam

Unsuitable extinguishing media for safety reasons:

carbon dioxide

Specific hazards:

harmful vapours

Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:

Wear a self-contained breathing apparatus.

Further information:

The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

6. Accidental Release Measures

Personal precautions:

Use personal protective clothing. Information regarding personal protective measures see, section 8.

Environmental precautions:

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:

For small amounts: Pick up with suitable appliance and dispose of.

For large amounts: Contain with dust binding material and dispose of.

Dispose of absorbed material in accordance with regulations.

7. Handling and Storage

Handling

Handle in accordance with good industrial hygiene and safety practice.

Protection against fire and explosion:

Avoid dust formation. Take precautionary measures against static discharges.

Storage

Suitable materials for containers: High density polyethylene (HDPE), Low density polyethylene (LDPE), Stainless steel 1.4541, Stainless steel 1.4571, Stainless steel 1.4439, Stainless steel 1.4301 (V2), Stainless steel 1.4401, Stainless steel 1.4539, Stainless steel 1.4361, Paper/Fibreboard
Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

8. Exposure controls and personal protection

Components with occupational exposure limits

sodium hydroxide, 1310-73-2; CLV 2 mg/m ³ (ACGIHTLV) TWA value 2 mg/m ³ (OEL (TH))
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Personal protective equipment

Respiratory protection:

Suitable respiratory protection for higher concentrations or long-term effect: Particle filter with low efficiency for solid particles (e.g. EN 143 or 149, Type P1 or FFP1)

Hand protection:

Chemical resistant protective gloves

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN 374):

(30513095/SDS_GEN_TH/EN)

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e.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), polyvinylchloride (0.7 mm) and other
 Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing. Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:

Wearing of closed work clothing is required additionally to the stated personal protection equipment. No eating, drinking, smoking or tobacco use at the place of work. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

Form:	granules	
Colour:	slightly yellowish	
Odour:	product specific	
Odour threshold:	No data available.	
pH value:	10.5 - 12.5 (10 g/l, 23 °C)	(DIN 19268)
pK _A :	10.6 (20 °C)	(OECD Guideline 112)
Melting point:	The substance / product decomposes therefore not determined.	
decomposition point:	> 300 °C	(DIN EN ISO 3146)
Boiling point:	not determined	
Flash point:	not applicable, the product is a solid	
Evaporation rate:	The product is a non-volatile solid.	
Flammability (solid/gas):	not flammable	
Upper explosion limit:	For solids not relevant for classification and labelling.	
Ignition temperature:	not determined	

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Self ignition:	not self-igniting	
Explosion hazard:	not explosive	
Fire promoting properties:	not fire-propagating	
Vapour pressure:	6 hPa (approx. 25 °C) contains water	
Bulk density:	650 - 900 kg/m ³	(DIN ISO 697)
Relative vapour density (air):	not determined	
Solubility in water:		
	approx. 600 g/l	
Miscibility with water:		
	soluble	
Hygroscopy:	Non-hygroscopic	
Partitioning coefficient n-octanol/water (log Pow):	<= -4 (25 °C; pH value: 11.4)	(Directive 92/69/EEC, A.8)
Surface tension:	71.5 mN/m (20 °C; 1 g/l)	(Directive 84/449/EEC, A.5, OECD harmonized ring method)
Viscosity, dynamic:	not applicable, the product is a solid	

Other Information:

If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Conditions to avoid:

See MSDS section 7 - Handling and storage.

Substances to avoid:

amphoteric metals, light metals

Corrosion to metals: Corrodes metals in the presence of water or moisture.

Hazardous reactions:

In spite of the dedusting carried out for reasons of industrial health a dust explosion hazard can not be excluded.

Hazardous decomposition products:

No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Acute toxicity

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Experimental/calculated data:

LD50 rat (oral): > 2,000 mg/kg (Guideline 92/69/EEC, B.1)

LC50 rat (by inhalation): > 5 mg/l 4 h

LD50 rat (dermal): > 2,000 mg/kg (OECD Guideline 402)

Irritation

Experimental/calculated data:

Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)

Serious eye damage/irritation rabbit: non-irritant (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:

Skin sensitizing effects were not observed in animal studies.

Experimental/calculated data:

Guinea pig maximization test guinea pig: Non-sensitizing. (OECD Guideline 406)

Germ cell mutagenicity

Assessment of mutagenicity:

The substance was not mutagenic in bacteria. The substance induced chromosomal aberrations in a mammalian cell culture test. At high concentrations the chelating properties interfere with the essential cation concentrations in the culture medium and within the cells. The substance was not mutagenic in a test with mammals.

Experimental/calculated data:

Ames-test

negative (OECD Guideline 471)

Ames-test

negative (OECD Guideline 472)

Cytogenetic assay

The genotoxic effect shown in vitro was not confirmed in vivo. (OECD Guideline 473)

HGPRT assay

negative (OECD Guideline 476)

Micronucleus assay

negative (OECD Guideline 474)

Carcinogenicity

Assessment of carcinogenicity:

In long-term studies in rats in which the substance was given by feed, a carcinogenic effect was not observed.

Experimental/calculated data:

rat (oral feed)

Result: negative

Reproductive toxicity

Assessment of reproduction toxicity:

The results were determined in a Screening test (OECD 421/422).

Developmental toxicity

Experimental/calculated data:

rat (oral feed)

NOAEL Teratog.: 1,000 mg/kg

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Experimental/calculated data:

Chronic Toxicity

rat oral feed (OECD Guideline 453)

NOAEL: 262 mg/kg

Aspiration hazard

| not applicable

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms. Based on long-term (chronic) toxicity study data, the product is very likely not harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations. Depending on local conditions and existing concentrations, disturbances in the biodegradation process of activated sludge are possible.

Toxicity to fish:

LC50 (96 h) > 100 mg/l, *Brachydanio rerio* (OECD 203; ISO 7346; 84/449/EEC, C.1)

Aquatic invertebrates:

EC50 (48 h) > 100 mg/l, *Daphnia magna* (OECD Guideline 202, part 1)

Aquatic plants:

EC50 (72 h) > 100 mg/l (growth rate), *Scenedesmus subspicatus* (Guideline 92/69/EEC, C.3, static)
Limited influence on algae growth due to chelate formation.

Microorganisms/Effect on activated sludge:

EC20 (0.5 h) > 1,000 mg/l, activated sludge, domestic (OECD Guideline 209, aerobic)

Chronic toxicity to fish:

No observed effect concentration (28 d) >= 100 mg/l, *Oncorhynchus mykiss* (OECD Guideline 204, Flow through.)

Nominal concentration.

Chronic toxicity to aquatic invertebrates:

No observed effect concentration, ≥ 100 mg/l, *Daphnia magna* (OECD Guideline 202, part 2)

Soil living organisms:

LC50 (14 d), *Eisenia foetida* (OECD Guideline 207, artificial soil)

The details of the toxic effect relate to the nominal concentration.

Terrestrial plants:

EC50 (19 d) 794 mg/kg, *Avena sativa* (OECD Guideline 208)

Mobility

Assessment transport between environmental compartments:

The substance will not evaporate into the atmosphere from the water surface.

Adsorption to solid soil phase is not expected.

Persistence and degradability

Assessment biodegradation and elimination (H₂O):

Readily biodegradable (according to OECD criteria).

Elimination information:

> 80 % BOD of COD (28 d) (OECD 301F; ISO 9408; 92/69/EEC, C.4-D) Readily biodegradable (according to OECD criteria).

80 - 90 % TIC of the ThIC (60 d) (OECD guideline 311) (anaerobic)

Bioaccumulation potential

Bioaccumulation potential:

No significant accumulation in organisms is expected as a result of the distribution coefficient of n-octanol/water (log Pow).

Additional information

Add. remarks environm. fate & pathway:

Treatment in biological waste water treatment plants has to be performed according to local and administrative regulations.

13. Disposal Considerations

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:

Uncontaminated packaging can be re-used.

Packs that cannot be cleaned should be disposed of in the same manner as the contents.

14. Transport Information

Domestic transport:

Not classified as a dangerous good under transport regulations

Sea transport
IMDG

Not classified as a dangerous good under transport regulations

Air transport
IATA/ICAO

Not classified as a dangerous good under transport regulations

15. Regulatory Information

Other regulations

16. Other Information

Information on intended use: This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use. This includes the mentioned and recommended usage. Any other intended applications should be discussed with the manufacturer. In particular this concerns the application for products that are the object of special standards and regulations.

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.